

SMALLER LOOKS ROUNDER: CORRECTING RESOLUTION-DEPENDENT BIAS OF SHAPE DESCRIPTORS IN BIOIMAGE AND RADIOMICS SOFTWARE

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ABSTRACT

Circularity and sphericity are ubiquitous in cell morphometry and radiomics, but the perimeter and surface estimates behind them depend on resolution. The default estimators of scikit-image (hence CellProfiler), OpenCV and ImageJ make the same object rounder the fewer pixels it covers, which we explain with a closed-form model; IBSI sphericity of a sphere drops from 0.92 to 0.82 as CT slices thicken from 1 to 5 mm. Even Crofton estimators fail for small objects and thick slices, as they miss chords shorter than the sampling step. We show that the missed chords number one sixth of the single-sample runs, yielding a parameter-free run-corrected Crofton estimator for 2D and anisotropic 3D grids. It is unbiased to within 0.005 on average down to 2-pixel radii and, for smooth lesions, up to 5 mm slices, five times more accurate than PyRadiomics on lung-tumour phantoms, and faster. On re-segmented nuclei, the default estimator shifts the form factor by up to 3.3 standard deviations and reverses a biological contrast at $10\times$; corrected estimators do not.

Index Terms— Morphometry, radiomics, harmonization, digital geometry, stereology

1. INTRODUCTION

Shape descriptors of segmented objects are a staple of quantitative imaging. Nuclear circularity quantifies pleomorphism in histopathology, CellProfiler’s *FormFactor* is part of every Cell Painting profile [1, 2], and sphericity is a standardized radiomics feature of the Image Biomarker Standardisation Initiative (IBSI) [3]. Both are ratios of a size to a boundary measure, $FF = 4\pi A/P^2$ and $\Psi = (36\pi V^2)^{1/3}/S$, equal to 1 for a disk and a ball. Studies routinely pool $20\times$ and $40\times$ slides or 1–5 mm CT slices.

The area or volume of a mask is easy to estimate, since the pixel count is unbiased under random placement; the boundary is not [4, 5]. Kulpa [4] derived the large-object length factor of chain codes; Crofton estimators, which count boundary crossings of lattice lines, are unbiased in the large-object limit [6], with error bounds due to the finite set of directions [7], and a recent comparison recommends them for 2D perimeters [8]; MATLAB patched its circularity for small objects in 2023 [9]. Three gaps remain: (a) no model explains

the *size dependence* of the estimators shipped in open-source tools, which is what breaks comparisons across resolutions; (b) Crofton estimators are also biased when objects are small or slices thick, because chords shorter than the sampling step along each line are never seen, an error distinct from the direction error and not corrected so far; (c) the consequences in real pipelines are unquantified.

Contributions. (i) A *run-corrected Crofton* (RCC) estimator: we show that, for smooth boundaries, the expected number of missed chords equals one sixth of the number of single-sample runs, which removes the small-object and thick-slice bias without parameters, in 2D and on anisotropic 3D grids. (ii) A closed-form model, $P \approx \kappa(L - 2\pi\delta)$, of the default estimators’ size dependence, and a post-hoc correction (F1) for existing feature tables. (iii) An audit of eight implementations against ground truth; evidence on 89k real nuclei (shifted features, distorted biological contrasts) and on 351 real CT reconstructions, where IBSI sphericity looks thickness-robust only through cancelling errors.

2. WHY DEFAULT ESTIMATORS DEPEND ON SIZE

Let $X \subset \mathbb{R}^2$ have a smooth boundary of length L , digitized on a unit grid as $D = X \cap \mathbb{Z}^2$ with random sub-pixel position and orientation. Most tools trace a polygon along D whose edges take a few lattice directions (Fig. 1a). If a boundary element ds at tangent angle θ becomes polygon length $g(\theta) ds$, and the polygon lies on average a distance δ inside ∂X , Steiner’s formula (a closed curve moved inward by δ has length $L - 2\pi\delta$) gives

$$P \approx \kappa(L - 2\pi\delta), \quad \kappa = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) d\theta. \quad (1)$$

Chain codes, used by scikit-image’s default `regionprops` perimeter (hence CellProfiler), OpenCV’s `arcLength` and MATLAB up to R2022b, join the centres of boundary pixels by steps of length 1 or $\sqrt{2}$: $g(\theta) = |\cos \theta| + (\sqrt{2} - 1)|\sin \theta|$ for $|\theta| \leq \pi/4$, so $\kappa = 8(\sqrt{2} - 1)/\pi \approx 1.055$ [4]. Along the grid axis closest to the normal, the last pixel centre inside X lies at a uniform distance in $[0, 1)$ from ∂X , i.e. $\cos \theta/2$ on average perpendicular to it, so $\delta = \frac{4}{\pi} \int_0^{\pi/4} \frac{\cos \theta}{2} d\theta = \sqrt{2}/\pi \approx 0.45$. For a disk of radius r pixels,

$$FF_{\text{chain}}(r) \approx \kappa^{-2} (r/(r - \sqrt{2}/\pi))^2, \quad (2)$$

which exceeds 1 for $r < 8.7$ px and tends to 0.899. *Marching squares* (PyRadiomics; IBSI meshes) have the same κ but $\delta = 0$, a constant factor κ^{-2} on FF; *pixel-edge polygons* have $\kappa = 4/\pi$, $\delta = 0$; ImageJ cuts the corners of pixel-edge outlines, a hybrid; MATLAB R2023a corrects δ but not κ [9]. In 3D, marching cubes on binary masks overestimates the surface increasingly with the slice-to-pixel ratio (Fig. 1e). *FI* (*post hoc*). Inverting Eq. (1), $\hat{P} = P_{\text{chain}}/\kappa + 2\pi\delta'$ repairs an existing scikit-image/CellProfiler perimeter column without the masks. Small objects (2–3 px) lose extra length, so we calibrate δ' once on the synthetic shapes (minimax FF error, giving 0.52) and use $\delta' = \frac{1}{2}$, i.e. $\hat{P} = P/1.055 + \pi$; no real data are used.

3. RUN-CORRECTED CROFTON ESTIMATOR

Crofton estimator. Consider lattice directions d (4 in 2D, 13 in 3D, in physical units for voxel spacing s , volume v). Lines along d through the pixel or voxel centres have density $\lambda_d = \|d\|/v$ per unit length or area orthogonal to d , and each line is sampled every $h_d = \|d\|$. If N_d is the number of runs (maximal sequences of object samples) along such lines, then N_d/λ_d estimates the object’s projection orthogonal to d (half the crossing measure for non-convex objects), and the Cauchy–Crofton formulas give

$$\hat{P} = \pi \sum_d w_d N_d / \lambda_d \text{ (2D)}, \quad \hat{S} = 4 \sum_d w_d N_d / \lambda_d \text{ (3D)}, \quad (3)$$

with w_d the fraction of directions (circle or sphere) closest to $\pm d$, computed for the actual spacing [6, 10].

Missed chords. A run is seen only if its chord contains a sample. With a uniformly random sample offset, a chord of length ℓ yields k samples with $P(k=0) = \max(0, 1 - \ell/h)$, and $P(k=1) = \ell/h$ for $\ell \leq h$, $2 - \ell/h$ for $h < \ell \leq 2h$. Short chords occur where a line grazes the boundary: at distance t from the tangent line of a boundary with curvature radius ρ , $\ell \approx 2\sqrt{2\rho t}$, and since t is uniform, the chord-length density is linear near zero, $f(\ell) \approx a\ell$. Hence the expected numbers of missed chords and of single-sample runs are

$$\mathbb{E}[\text{missed}_d] = \int_0^h a\ell \left(1 - \frac{\ell}{h}\right) d\ell = \frac{ah^2}{6},$$

$$\mathbb{E}[n_{1,d}] = \int_0^h a\ell \frac{\ell}{h} d\ell + \int_h^{2h} a\ell \left(2 - \frac{\ell}{h}\right) d\ell = ah^2, \quad (4)$$

so, whatever a and h , the missed chords are one sixth of the single-sample runs:

$$N_d^{\text{RCC}} = N_d + \frac{1}{6} n_{1,d}, \quad (5)$$

where $n_{1,d}$ counts runs of exactly one sample along d . Plugging N_d^{RCC} into Eq. (3) gives the RCC estimator. It needs no parameter or resampling; anisotropy enters only through h_d .

4. EXPERIMENTS

Synthetic ground truth. 5,000 shapes of known perimeter and area: ellipses (axis ratios 1, 0.6, 0.35) and two families

of random smooth star-shaped “nuclei”, equivalent radii 2–48 px, random rotation and sub-pixel offset. In 3D, randomly rotated spheres and 1:0.75:0.5 ellipsoids (equivalent radius 3–20 mm) on $1 \times 1 \times s$ mm grids, $s \in [1, 5]$. We run the libraries’ own code (scikit-image 0.26 [11], OpenCV 4.13, PyRadiomics 3.0.1 [12]) and CellProfiler’s measurement code; ImageJ’s [13] traced perimeter is ported, MATLAB R2023a’s formula [9] implemented.

Real nuclei. PanNuke fold 1 [14] (H&E, 40 $\times$, annotated types; 43,519 nuclei) and BBBC038 [15] (22,858 nuclei). Lower magnification is obtained by *re-digitizing* each annotated mask on a 2 $\times$ or 4 $\times$ coarser grid (block mean ≥ 0.5), and, realistically, by *re-segmenting* block-averaged images independently at each resolution with the pretrained Cellpose [16] nuclei model, matching nuclei by IoU ≥ 0.5 (12,279 PanNuke and 10,454 BBBC038 nuclei). We report the paired shift Δ/σ (mean change in FF over its native s.d.; s.e. ≈ 0.01), Spearman’s ρ_A between area and FF, Cohen’s d between PanNuke nucleus types, and the 10 $\times$ balanced-accuracy drop of a 40 $\times$ -trained logistic regression on shape features.

Lung tumours. The 63 CTs of the Medical Segmentation Decathlon lung set [17] provide real tumour shapes; each tumour (≥ 300 mm³; 64 tumours, radius 3–45 mm) is turned into a phantom: the mask is resampled to 0.5 mm and smoothed ($\sigma = 1.5$ mm), its 0.5 level set defines a surface of known sphericity (0.60–0.96, from a fine mesh), point-sampled at 0.7 mm in-plane with 0.7–5 mm slices. Real reconstructions come from RIDER Lung CT [18]: 32 patients scanned twice on the same day, each scan reconstructed at 1.25, 2.5 and 5 mm with a sharp and a smooth kernel, with radiologist tumour segmentations for every reconstruction (351 series with uniform slice spacing).

5. RESULTS

Synthetic 2D (Fig. 1b). The chain-code tools overestimate circularity by +0.55 at $r = 2$ px and +0.11 at $r = 4$ px and underestimate it by 0.08 at $r = 48$, as Eq. (2) predicts (dashed). ImageJ follows the same trend (+0.09 to –0.08); PyRadiomics and MATLAB R2023a are offset by κ^{-2} (–0.08 to –0.09). Crofton is unbiased for large objects but overestimates circularity by 0.050 at $r = 2$ and 0.013 at $r = 4$ px. RCC reduces this to ≤ 0.005 at every size, at similar variance; for the most irregular shapes it halves the bias at $r = 2$ (0.10 $\rightarrow$ 0.05).

Synthetic 3D (Fig. 1e). PyRadiomics sphericity is biased by –0.08 at 1 mm and –0.17 at 5 mm slices for every lesion size; isotropic resampling, as IBSI recommends, *increases* the bias to –0.19. RCC stays within ± 0.005 for all $R \geq 5$ mm and thicknesses (Crofton: up to +0.046). On the 64 (irregular) lung-tumour phantoms, PyRadiomics is biased by –0.065 at 0.7 mm and –0.136 at 5 mm slices, Crofton by +0.005 and +0.042, and RCC by +0.002 and +0.026, a fivefold smaller

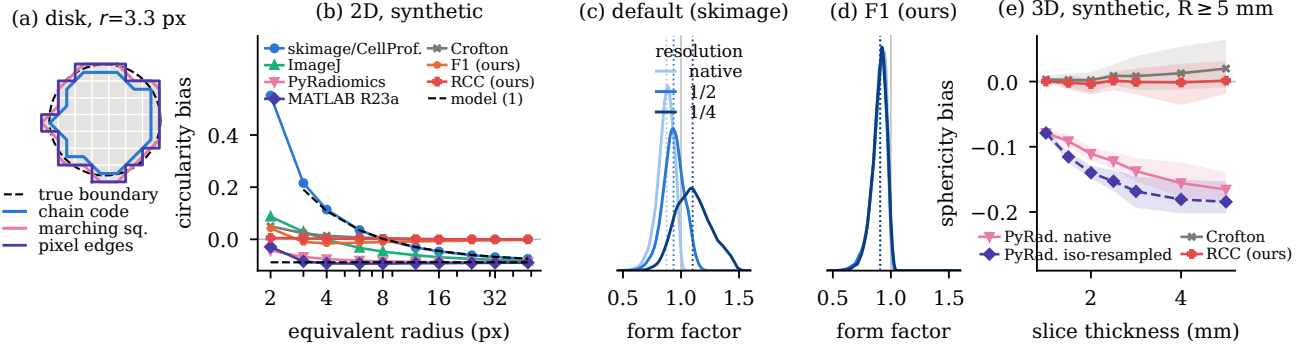


Fig. 1. (a) Traced polygons: chain codes run inside the boundary ($\delta > 0$), marching squares on it, pixel edges outside. (b) Circularity bias on 5,000 synthetic shapes; dashed: Eq. (1). (c,d) FF of 10,454 BBBC038 nuclei re-segmented by Cellpose at native, $\frac{1}{2}$, $\frac{1}{4}$ resolution (dotted: medians), default estimator (c) vs. F1 (d). (e) Sphericity bias of rotated spheres/ellipsoids ($R \geq 5$ mm); bands: 10–90%.

Table 1. Real nuclei: paired shift Δ/σ of FF when the resolution is halved ($\frac{1}{2}$) or quartered ($\frac{1}{4}$), by re-digitization of the annotations (“re-dig.”) or independent Cellpose re-segmentation (“re-seg.”), and area–FF correlation ρ_A at native resolution. Ideal: zero shift.

FF from	PanNuke (H&E)					BBBC038 (fluorescence/brightfield)				
	re-dig.		re-seg.		ρ_A	re-dig.		re-seg.		ρ_A
	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$		$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	
scikit-image / CellProfiler	+0.72	+3.25	+0.68	+2.53	−0.34	+0.92	+5.56	+0.95	+3.30	−0.56
MATLAB R2023a	+0.06	+0.27	0.00	+0.16	−0.05	+0.11	+0.51	+0.04	+0.20	−0.19
Crofton	+0.11	+0.31	+0.06	+0.25	−0.09	+0.16	+0.51	+0.13	+0.46	−0.24
F1 (ours, post hoc)	+0.03	+0.16	−0.03	+0.05	−0.03	+0.07	+0.35	−0.01	+0.05	−0.16
RCC (ours)	+0.09	+0.10	+0.03	+0.09	−0.08	+0.11	+0.12	+0.10	+0.28	−0.22

Table 2. PanNuke: Cohen’s d of FF vs. neoplastic nuclei (s.e. ≈ 0.014); “mixed”: inflammatory at $20\times$ vs. neoplastic at $40\times$; drop: $10\times$ balanced-accuracy loss of a $40\times$ -trained type classifier (no FF: 0.008).

FF from	infl.-neo.		conn.-neo.		mixed	drop
	$40\times$	$10\times$	$40\times$	$10\times$		
scikit-image / CellProfiler	0.67	0.32	−0.62	+0.11	1.62	0.115
PyRadiomics, Crofton	0.40	0.53	−0.76	−0.62	0.56	0.041
F1 (ours)	0.35	0.39	−0.79	−0.41	0.35	0.054
RCC (ours)	0.38	0.31	−0.77	−0.72	0.49	0.031

error (residual: poles thinner than a slice, irregular boundaries).

Real reconstructions (RIDER). Test–retest repeatability is the same for all estimators (ICC 0.81–0.94 per reconstruction), so RCC costs no precision. From 1.25 to 5 mm slices, RCC sphericity of the same tumours rises by $+0.097/+0.098$ (sharp/smooth kernel; $p < 10^{-8}$, Wilcoxon, $n = 55$ –58) because thick-slice images and contours are smoother. PyRadiomics hides this ($+0.009$, $+0.010$, n.s.) only because its own bias falls with thickness: the PyRadiomics–RCC gap

changes by -0.088 , matching the -0.078 predicted by the phantoms. An apparently “robust” IBSI sphericity thus results from two cancelling errors, not from invariance.

Real nuclei (Fig. 1c,d; Table 1). With the default estimator, halving the resolution shifts the mean FF by 0.7 – 0.95σ and quartering it by 2.5 – 5.6σ in both settings and modalities; at $\frac{1}{4}$ resolution the median BBBC038 nucleus has an impossible $FF > 1$. The default also creates an area–roundness correlation ($\rho_A = -0.34$, -0.56). Crofton reduces the shifts to 0.06 – 0.51σ and RCC to 0.03 – 0.28σ ; F1 is the most stable under re-segmentation ($\leq 0.05\sigma$). **Biological contrasts (Table 2).** Inflammatory nuclei are smaller than neoplastic ones (median radius 9.5 vs. 13.9 px at $40\times$), so the default inflates their extra roundness from $d = 0.35$ – 0.40 to 0.67. At $10\times$, the default reverses the sign of the connective–neoplastic contrast ($-0.62 \rightarrow +0.11$), whereas RCC preserves it ($-0.77 \rightarrow -0.72$). When scanners are mixed (inflammatory at $20\times$, neoplastic at $40\times$), the default reports $d = 1.62$ instead of 0.35 – 0.49 . A $40\times$ -trained type classifier loses 0.115 balanced accuracy at $10\times$ with the default FF and 0.031 with RCC.

6. DISCUSSION

The default circularity of scikit-image, CellProfiler and OpenCV distorts any comparison across pixel sizes (Eq. (2)); constant-offset estimators mislead absolute thresholds and, in 3D, can mask real protocol effects. Both fixes are cheap. RCC takes <1 ms per 2D object and 3–15 ms per 3D lesion, $9\text{--}90\times$ faster than PyRadiomics’ mesh for lesions of radius 15–40 mm. F1 needs no images: we ran CellProfiler’s own measurement code on the 68,574 BBBC038 masks (its perimeter equals scikit-image’s to machine precision) and repaired the resulting table in 4 ms, reproducing Table 1. We recommend RCC for new analyses on the native grid, F1 for existing CellProfiler/scikit-image tables, and reporting object sizes in pixels. Limitations: Eq. (5) assumes smooth boundaries (objects of about one slice remain biased), and block-averaging only approximates re-imaging. Code: <https://github.com/Hannibal0319/isbi2027-shape-bias>.

7. COMPLIANCE WITH ETHICAL STANDARDS

Only public anonymized data were used; no ethics approval was required.

8. ACKNOWLEDGMENTS

No funding or conflicts of interest. The AI assistant Claude (Anthropic) helped write code and text.

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